

Schema-Aware Grounding Effects in PostGIS Natural-Language-to-SQL: A Subset-Decomposed Evaluation Across Eleven LLMs

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Abstract

Natural-language interfaces to spatial databases increasingly use schema-aware grounding: prompt-side evidence about table aliases, geometry columns, spatial reference systems, PostGIS operators, value hints, and safety constraints. For computational geoscience applications, this is a spatial-database informatics problem rather than only a language-model benchmarking problem. We evaluate grounding as an intervention on CQ-125, a 125-question Chongqing PostGIS benchmark containing 85 spatial-query questions and 40 robustness questions, across 11 LLM families with $N=3$ paired samples per family. All paired tests use question-level majority vote across the stochastic samples followed by a single exact two-sided McNemar test, avoiding the pseudoreplication that results from treating NQ repeated samples as independent observations. Grounding produces a subset-dependent pattern. On the Robustness subset, all 11 families improve in point estimate (+12.50 to +65.00 pp; nine significant in unadjusted per-cell tests and after Holm correction within the Robustness comparison set, one marginal, one non-significant positive). On the Spatial subset, 7 families show unadjusted significant positive gains (+11.76 to +23.53 pp), 2 show marginal positive gains, and 2 show non-significant negative point estimates; Holm correction within the Spatial comparison set leaves 3 positive cells significant. The most negative panel estimate is gemini-3.5-flash ($\Delta = -10.59$ pp, $p = 0.136$), and a balanced $N=5$ focused check on the same family retains this negative but non-significant Spatial point estimate. We therefore make a bounded claim: subset decomposition and de-pooled paired testing are necessary for defensible grounding-effect evaluation in geospatial NL2SQL; cross-family effects are heterogeneous; and the observed negative tail requires out-of-benchmark replication before it can be treated as a stable model-family property.

Keywords: NL2GeoSQL; PostGIS; semantic grounding; spatial-SQL execution accuracy; LLM evaluation; subset analysis; paired statistical testing; large language model.

1 Introduction

Natural language access to spatial databases is a long-standing GIScience problem (Goodchild, 1992; Egenhofer and Franzosa, 1991; Clementini et al., 1993). Recent large language models have advanced general text-to-SQL on benchmarks such as Spider (Yu et al., 2018) and BIRD (Li et al., 2023); PostGIS-based NL2GeoSQL adds the OGC Simple Feature Access predicates (Open Geospatial Consortium, 2010), coordinate-reference-system transformations, the geometry/geography type distinction, KNN operators, and the safety constraints any production GIS must enforce (PostGIS Development Team, 2024). These requirements make geospatial text-to-SQL a GIScience problem

as well as a language-model problem: the system must translate geographic concepts into executable spatial predicates while preserving units, coordinate reference assumptions, result shape, and read-only behaviour.

The standard engineering response is schema-aware grounding: an upstream stage that injects table aliases, value enumerations, geometry-type metadata, SRID flags, large-table boundedness rules, and PostGIS dialect cues into the LLM prompt. Grounding is often evaluated as a single positive intervention, but this aggregate view is too coarse for mixed geospatial benchmarks. A benchmark that combines ordinary spatial queries with robustness cases is not testing one homogeneous capability. Spatial questions reward precise predicate choice, join structure, geometry measurement, and output-shape control; robustness questions reward refusal, schema enforcement, and bounded execution. The same grounding context can therefore interact differently with each subset.

This paper evaluates that interaction on CQ-125, a 125-question Chongqing PostGIS benchmark containing 85 spatial-query questions and 40 robustness questions. Across 11 LLM families at $N=3$ paired samples per family, grounding has uniformly positive point estimates on Robustness (+12.50 to +65.00 pp). On Spatial, however, effects are heterogeneous: seven families improve in unadjusted per-cell tests, two are marginal positive, and two have non-significant negative point estimates; after Holm correction within the 11 Spatial tests, three positive cells remain significant. The most negative panel estimate is gemini-3.5-flash (-10.59 pp, $p = 0.136$). A balanced $N=5$ three-condition analysis on the same family, using no-grounding, grounding-only, and grounding plus a minimal prompt patch, retains the same Spatial grounding-only point estimate (-10.59 pp, $b/c = 19/10$, $p = 0.136$) and a large Robustness gain (+27.50 pp, $p = 0.001$). The stable conclusion is therefore not that grounding generally hurts. It is that grounding effects are subset-conditional and family-conditional, and that aggregate execution accuracy can hide the interaction.

Contributions.

1. Methodological (§6.6). Aggregate execution accuracy is insufficient for grounding-effect claims on geospatial NL2SQL when robustness and spatial questions are mixed. The paired-test convention also matters: pooling N stochastic samples $\times Q$ questions into a single McNemar test on NQ observations violates independence and inflates significance. We use question-level majority vote across N samples followed by a single exact two-sided McNemar test on the per-question table.
2. Empirical (§6). On 11 LLM families surveyed at $N=3$ on CQ-125, grounding improves Robustness in all families by point estimate, but Spatial effects range from -10.59 to $+23.53$ pp. This establishes cross-family heterogeneity rather than a universal positive or negative effect.
3. Diagnostic and design implication (§6.4–§6.5). For gemini-3.5-flash, a focused three-condition analysis identifies a candidate negative-tail case and links many regressions to unsolicited SQL elaboration. A perfect-classifier routing calculation gives an upper bound for subset-conditional grounding, but we treat it as a design implication, not a validated production policy.

The semantic-layer framework, intent-conditioned router, SQL postprocessor, and execution-time self-correction loop on which grounding is implemented are described in §3. The new questions are not how to ground, but when grounding helps which question class on which family and how to test that question with an honest paired statistic. As supporting deployment context, not as primary inferential evidence, §5.2 reports a fixed-host Gemma4 model-size sweep.

The paper proceeds as follows. §2 reviews GIScience foundations and recent NL-to-spatial-database systems. §3 summarises the framework. §4 describes the experimental setup. §5 reports

framework behaviour and bridges to §6, the central empirical section. §7 discusses interpretation, limitations, and ties to GIScience. §8 concludes.

2 Related Work

2.1 GIScience foundations for NL-to-spatial-database

Natural language access to spatial databases sits within the GIScience programme articulated by Goodchild (1992): formalising geographic concepts so they can be computed rigorously. The relational spatial semantics we ground against have roots in the 4- and 9-intersection formalisms (Egenhofer and Franzosa, 1991) and the simplified end-user topological relations of Clementini et al. (1993) — equals, disjoint, touches, contains, overlaps, within, crosses, intersects — which the OGC Simple Feature Access standard codifies for relational spatial SQL (Open Geospatial Consortium, 2010) and GeoSPARQL (Open Geospatial Consortium, 2012) extends to RDF. PostGIS (PostGIS Development Team, 2024) operationalises these in PostgreSQL and defines the geometry/geography type distinction, the `::geography` cast, `ST_DWithin`, `ST_Transform`, and the `<->` KNN operator any PostGIS-generating system must respect. Our grounding rules encode these formalisms as prompt-side evidence.

2.2 Natural language interfaces to spatial databases

Early geospatial question answering focused on small fact databases: Zelle and Mooney (1996) used inductive logic programming for GeoQuery, and Punjani et al. (2018) extended this to template-based question answering over linked geospatial data, highlighting spatial-predicate coverage and coordinate-system awareness. Recent LLM-era work falls into two groups. The first targets end-to-end NL-to-executable-spatial-SQL: NALSpatial (Liu et al., 2025b) translates user queries into spatial SQL (implemented over the SECONDO spatial DBMS rather than PostGIS); Monkuu (Yu et al., 2025) is an IJGIS-published LLM-powered NL interface with dynamic schema mapping; GeoSQL-Eval (Hou et al., 2026) contributes a large-scale LLM evaluation on PostGIS functions and NL2GeoSQL tasks. The second targets higher-level GeoAI agents: GeoCogent (Hou et al., 2025) is an LLM-based agent for geospatial code generation; GeoAgent (Lin et al., 2026) is a hierarchical multi-agent architecture. Surveys map the broader opportunities (Mai et al., 2024).

Our contribution is narrower than a general GeoAI agent and different from a function-coverage benchmark. We study grounding as an intervention in end-to-end execution-graded NL-to-PostGIS-SQL, where the semantic layer explicitly encodes OGC predicates, SRIDs, geography casts, KNN operators, and read-only bounded-output constraints. The central question is not whether a system can use PostGIS functions in isolation, but how prompt-side spatial evidence changes executable SQL correctness across spatial-query and robustness subsets. Compared with NALSpatial and Monkuu, we contribute a 125-question execution benchmark with an explicit Robustness suite, paired question-level McNemar testing, and an 11-family grounding-effect panel. Compared with GeoSQL-Eval, the benchmark is smaller but is designed to support paired intervention analysis rather than broad function inventory.

Table 1 summarises the relevant distinction: our benchmark is not the largest PostGIS inventory, but it supports paired grounding-intervention analysis with a robustness suite and released evaluation records.

Table 1: Direct comparison with recent NL-to-spatial-database systems. NALSpatial is implemented over SECONDO, not PostGIS. ✓: supported / reported; —: not reported.

System	E2E	Function	Safety	Paired gnd.	Released
Our framework	✓	—	✓	✓	✓
NALSpatial (Liu et al., 2025b)	—	—	—	—	✓
Monkuu (Yu et al., 2025)	✓	—	—	—	—
GeoSQL-Eval (Hou et al., 2026)	—	✓	—	—	✓
GeoCogent (Hou et al., 2025)	—	—	—	—	—
GeoAgent (Lin et al., 2026)	—	—	—	—	—

2.3 Text-to-SQL with LLMs and semantic layers

LLM text-to-SQL has progressed through Spider (Yu et al., 2018; Rajkumar et al., 2022), BIRD (Li et al., 2023), Spider 2.0 (Lei et al., 2025), and BEAVER (Chen et al., 2024). DIN-SQL (Pourreza and Rafiei, 2023) introduced decomposed in-context prompting with schema linking, query classification, generation, and self-correction; DAIL-SQL (Gao et al., 2024) introduced skeleton-similarity few-shot selection; C3 (Dong et al., 2023) explored zero-shot ChatGPT text-to-SQL with calibrated bias correction; CSpider (Min et al., 2019) is an early multilingual benchmark. Semantic-layer thinking — entities, measures, dimensions, metric graphs — is operationalised in tools such as MetricFlow (dbt Labs, 2026), which encodes metric definitions in dbt-based analytical warehouses; we adapt the same entity/measure/dimension vocabulary by combining it with GIS annotations (geometry types, SRIDs, OGC predicates). Here BIRD serves as a warehouse track of the same bilingual framework, reported as a design/held-out pair so the warehouse path is measurable and the next engineering round scopable.

3 Method: Semantic-Layer Framework

3.1 Formal scope: semantic layer as metadata graph $G = (V, E)$

We formalise the semantic layer as a directed multigraph $G = (V, E)$ that encodes the executable subset of OGC Simple Feature Access (Open Geospatial Consortium, 2010) required for PostGIS grounding. $V = E_{geo} \cup D \cup M \cup I \cup P \cup C$ where E_{geo} are geographic-entity (PostGIS table) vertices with attributes (geometry column, SRID, geometry type); D, M are dimension/measure vertices for the warehouse track; I are intent tags (attribute_filter, aggregation, spatial_join, spatial_measurement, knn, preview_listing); P are OGC SFA predicate vertices partitioned into topological, metric, and KNN classes; C are constraint vertices for SRID ranges, unit rules, and safety/refusal. E comprises $E_{hasGeom}, E_{fk}, E_{topo}, E_{metric}, E_{knn}, E_{route}, E_{safety}, E_{unit}$ (table-geometry, foreign keys, predicate-entity edges, intent→predicate routing, refusal edges, and SRID range→required cast→yielded unit chains). Algorithm 1 materialises G at agent startup from the live PostGIS catalog. G is an executable subset of the OGC SFA predicate taxonomy, not a complete geospatial ontology; full GeoSPARQL/ISO 19107 alignment is future work, and Table 2 documents the rule-level mapping we currently enforce.

3.2 Problem formulation

We formulate cross-domain natural language to SQL as the task of translating a user question q into an executable SQL statement s over heterogeneous structured databases that may belong to either geospatial or conventional warehouse domains. Let S denote the executable schema representation

Algorithm 1 BuildSemanticGraph(*schema*, *prefix*)

```

1:  $G \leftarrow$  empty multigraph
2: for  $(t, c, srid, \tau) \in \text{geometry\_columns}$  do
3:   if  $t$  starts with prefix and  $srid \neq 0$  then
4:     add GEO_ENTITY vertex  $(t, c, srid, \tau)$ 
5:   end if
6: end for
7: for  $(t_s, t_t) \in \text{key\_column\_usage}$  do
8:   add edge  $(t_s \rightarrow t_t, \text{FK})$ 
9: end for
10: register topological / metric / KNN predicate vertices (OGC SFA 06-104r4)
11: register intent vertices; route intents  $\rightarrow$  predicate classes
12: register unit-rule edges: metric  $\rightarrow$  SRID range  $\rightarrow$  required cast
13: return  $G$ 

```

Table 2: Alignment of semantic-layer rules to OGC SFA (Open Geospatial Consortium, 2010), ISO 19107 (ISO/TC 211, 2019) clauses, and PostGIS functions (PostGIS Development Team, 2024). Rule-level mapping; each enforced rule traces to a specific clause.

Rule	Edge class	OGC 06-104r4 clause	ISO 19107 clause	PostGIS
R_topo_intersects	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	ST_Intersects
R_topo_contains	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	ST_Contains
R_topo_within	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	ST_Within
R_metric_length_geog	E_{unit}	§6.1.2.3 (Length) + §8.2.3	§6.4.14	ST_Length(geog)
R_metric_area_geog	E_{unit}	§6.1.2.3 (Area) + §8.2.3	§6.4.15	ST_Area(geog)
R_metric_dwithin_geog	E_{unit}	§6.1.2.3 (Distance) + §8.2.3	§6.4.10	ST_DWithin(geog)
R_knn_nearest	E_{knn}	§6.1.2.3 (Distance)	§6.4.10	<-> KNN
R_safety_refusal	E_{safety}	not in OGC	—	postprocessor AST
R_safety_bounded	E_{safety}	not in OGC	—	LIMIT injection

extracted from the database catalog, and let E denote semantic-side evidence such as aliases, hierarchy expansions, geometry metadata, value hints, or fact/dimension/entity annotations. The problem can then be written abstractly as

$$\hat{s} = f(q, S, E), \quad (1)$$

where f is implemented by an LLM-conditioned generation pipeline rather than a single end-to-end model. Unlike standard text-to-SQL settings, the target databases in our problem formulation are not assumed to share homogeneous operator semantics. In warehouse-style databases, correct SQL generation primarily depends on schema linking, join reasoning, aggregation, and temporal filtering. In geospatial databases, however, the model must additionally reason about geometry-bearing columns, spatial relations, coordinate systems, and domain-specific operators such as intersection, containment, buffering, and area or distance calculation. The purpose of our framework is therefore not only to generate valid SQL, but also to provide a semantic mediation layer that adapts the grounding process to the structural characteristics of each domain.

3.3 Three-stage pipeline

We design NL2Semantic2SQL as a three-stage pipeline (Figure 1). In the first stage, the system performs semantic resolution over a semantic layer that stores table-level and column-level anno-

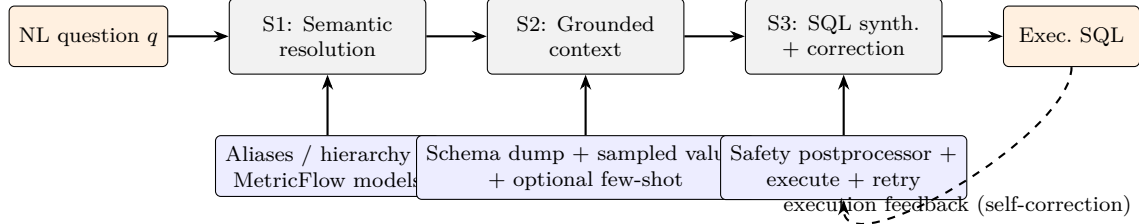


Figure 1: Architecture of NL2Semantic2SQL: Stage 1 semantic resolution, Stage 2 grounded-context assembly, Stage 3 SQL synthesis with postprocessor and self-correction.

tations, aliases, domain hints, and task-relevant metadata. In the second stage, the resolved information is combined with database schema inspection and optional few-shot retrieval to construct a grounded context block for SQL generation. In the third stage, the generated SQL is postprocessed, executed under safety constraints, and, when necessary, revised through an execution-feedback self-correction loop. This decomposition separates semantic interpretation from SQL synthesis, allowing the framework to expose domain-specific evidence to the language model while keeping the final generation step constrained by executable schema information.

3.4 Geospatial-aware grounding and bilingual routing

The framework’s geospatial path explicitly identifies geometry columns, geometry types, and spatial reference information, and injects operator-level rules into the grounding context. These rules distinguish topological from metric predicates and specify when area or distance calculations require coordinate transformation or geography casting, reducing the burden on the language model to infer execution constraints from raw schema text. When spatial signals are absent, the framework falls back to generic schema linking and aggregation-oriented prompt construction. Routing between these paths is driven by a bilingual intent classifier that recognises Chinese GIS keywords (spatial operator terms, Chinese administrative region names, Chinese measurement units) alongside English warehouse expressions (“how many”, “average”, “compare”). In real-world deployments GIS analysts often query in Chinese while warehouse analysts use English, and a single interface must handle both without language-specific model swapping; bilingual routing is therefore a first-class design choice rather than an afterthought.

3.5 Warehouse-side enhancements

For non-geospatial warehouse queries, the framework adds three domain-adaptive enhancements. Cross-domain candidate ranking reprioritises sources based on whether the parsed semantic context contains spatial operators or region filters; when such signals are absent, geometry-bearing tables are deprioritised and tables with matched column evidence are boosted. Value-aware grounding injects sample distinct values from low-cardinality text columns so that the model can use exact categorical values rather than hallucinated strings. MetricFlow-style fact/dimension modelling stores per-table roles, entity columns (primary or foreign join keys), and measure columns in a semantic-model registry; at grounding time, candidate tables are matched against this registry and explicit join-path hints are derived by aligning shared entities across fact/dimension pairs. These hints are injected only for non-spatial queries, leaving the GIS path unchanged.

3.6 Execution-time self-correction

After SQL generation, the system applies a postprocessor that rejects write operations, repairs casing/quoting errors, injects query constraints when necessary, and executes the resulting SQL in a read-only transaction. If execution fails, the framework enters a bounded self-correction loop. Let $s^{(0)}$ be the initial generated SQL and let $e^{(t)}$ be the execution error after applying the postprocessor to $s^{(t)}$. The retry policy is:

$$s^{(t+1)} = g(q, s^{(t)}, e^{(t)}, S), \quad t = 0, 1, \dots, T-1, \quad (2)$$

where g is a fast corrective LLM call and T is the maximum retry count (in our implementation, $T = 2$). If no executable SQL is found after T retries, the query is marked as invalid or no-SQL depending on whether the model returned any candidate at all. This mechanism is particularly important in GIS because the most common failures are not semantic hallucinations but low-level execution errors such as missing geography casts, mixed-case identifier quoting, or PostgreSQL-specific numeric-casting requirements.

3.7 Worked example: end-to-end PostGIS grounding

A worked example tracing a Chongqing benchmark question (length-of-one-way-roads in km) through every pipeline stage — including the `::geography` cast injected by Stage 2, the unit-conversion rule, the postprocessor’s row-shape sanity check, and the resulting execution-based match against the gold SQL — is provided in Supplement S2. Three implementation artefacts support the flow: a per-database YAML semantic layer (table aliases, geometry columns with SRID, rule identifiers); a small intent-rule set (`attribute_filter`, `aggregation`, `spatial_join`, `spatial_measurement`, `knn`, `preview_listing`, each associated with a minimal set of prompt-side rules); and the postprocessor (write-node rejection, identifier repair against the real schema, large-table LIMIT injection, the execution-time retry loop above). The anonymised companion repository releases minimal versions of all three.

3.8 Multi-track evaluation protocol

Evaluation is performed on three benchmark tracks. The geospatial track uses the 125-question GIS benchmark (85 Spatial + 40 Robustness), covering spatial selection, overlap analysis, metric distance queries, buffering, spatial aggregation, and a dedicated robustness suite that probes refusal behaviour on dangerous, illusory, schema-mismatched, and out-of-memory requests. The warehouse track uses BIRD mini_dev (Li et al., 2023) with two disjoint subsets: a 108-question design set and a 150-question independent held-out set, both evaluated in single-pass mode. The cross-lingual track pairs 50 Chinese-translated BIRD questions with 50 native-Chinese GIS questions. For all tracks we use execution accuracy (EX) as the primary metric: a prediction is correct if and only if its execution result set matches the gold SQL under set-equality with numeric tolerance. We additionally report 95% Wilson confidence intervals, execution validity, per-difficulty breakdowns, paired McNemar tests (two-sided exact binomial), and token cost. Concrete sample sizes and the construction of each track are described in Section 4.

Table 3: Evaluation-protocol inventory. MV denotes question-level majority vote before paired McNemar testing. A is no-grounding, B is grounding-only, and C is grounding plus the minimal-modification prompt profile.

Experiment		Dataset / model scope	Conditions and samples	Role in the manuscript
Cross-family panel	grounding	CQ-125; 11 LLM families	A/B, $N=3$, MV McNemar	Primary evidence for subset-conditional and family-conditional grounding effects.
Focused Gemini 3.5 diagnostic	diag-	CQ-125; gemini-3.5-flash	A/B/C, balanced $N=5$, MV McNemar	Diagnostic check of the negative Spatial tail and mini-mod candidate patch.
Framework context		GIS, BIRD, cross-lingual, and DIN-SQL tracks	Original paired protocols; BIRD in single-pass mode	Supporting system context, not the primary grounding-effect claim.
Gemma4 fixed-host sweep		CQ-125; five local Gemma4 scales on host228	Single-host frozen summaries; best-observed full rows where multiple full runs existed	Deployment accuracy-runtime probe only.
FloodSQL-Bench audit	smoke	Local PostGIS port; gemini-2.5-flash	Stratified 150q, $N=1$	External harness-readiness audit only.

4 Experimental Setup

4.1 Experimental setup

Evaluation protocols. The primary analysis in this manuscript is the CQ-125 grounding-effect study in §6. It compares two prompt conditions, no-grounding (A) and grounding-only (B), across 11 LLM families using $N=3$ paired samples per family and question-level majority-vote McNemar tests. A focused gemini-3.5-flash diagnostic adds a third condition, grounding plus minimal-modification rules (C), and is run as a balanced A/B/C comparison with $N=5$ samples per condition under the same question-level majority-vote convention. Table 3 separates the inferential panel from supporting framework, deployment, and harness-readiness results.

The framework-level results in §5 use a separate evaluation protocol and are reported as supporting context rather than as the primary claim. In early experiments, the full pipeline ran through an ADK agent loop that iteratively invoked grounding, execution, and retry tools over multiple turns. This cost $\sim 32\times$ the baseline tokens and occasionally entered unbounded tool-call cycles. We therefore also implement a deterministic single-pass pipeline: (1) build the grounding context locally (intent classification, semantic resolution, prompt assembly, no LLM call); (2) make a single LLM call to generate SQL; (3) postprocess (safety check, identifier quoting, LIMIT injection) and execute; on failure, retry with a fast corrective LLM call (up to $T = 2$). Single-pass mode reduced BIRD token cost from $\sim 32\times$ to $7.9\times$ and improved design-set EX. BIRD results below use single-pass; the earlier GIS framework headline uses the agent loop and is separated from the cross-family grounding-effect analysis.

We also include a supporting Gemma4 model-size sweep on the same local Ollama serving node (host228). This sweep evaluates five Gemma4 parameter scales (e2b, e4b, 12b, 26b, and 31b) on CQ-125 under the same baseline-vs-full framework configurations and records wall-clock runtime. It is a single-host engineering probe, not part of the $N=3$ majority-vote McNemar panel in §6; accordingly, we use it only to document a practical accuracy-runtime trade-off. The source-table summary is distributed with the manuscript source and is excluded from the primary statistical

Table 4: CQ-125 Spatial-subset profile. Predicate families may overlap because one question can require more than one spatial operation. The 40-question Robustness subset is reported separately.

Category	Count
Predicate family (a query may belong to several)	
topological	21
metric	20
knn	5
aggregation	46
plain	23
PostGIS function frequency	
ST_Transform	14
ST_Intersects	12
ST_Area	7
ST_Length	6
ST_Contains	6
ST_Distance	5
<->	5
ST_AsText	3
ST_Within	3
ST_Centroid	2
ST_DWithin	2
ST_Union	2
Multi-table join multiplicity	
single-table	55
2-table join	29
3+ table join	1
Difficulty stratification	
Easy	24
Medium	36
Hard	25
Total Spatial questions	85

reductions.

Benchmarks. We evaluate under three tracks. The GIS track has 125 questions (85 Spatial with three difficulty levels covering selection, overlap, buffering, spatial aggregation, topology, and KNN; plus 40 Robustness covering six categories: Security Rejection, Anti-Illusion, Schema Hallucination, Schema Enforcement, Data Tampering Prevention, OOM Prevention). The BIRD track uses BIRD mini_dev (Li et al., 2023) with a 108-question design set (used to derive grounding rules by error attribution) and a disjoint 150-question held-out set sampled from the remaining BIRD mini_dev PostgreSQL items with proportional difficulty stratification. The cross-lingual track has 50 Chinese-translated BIRD questions (drawn from the 108 design) plus 50 native Chinese GIS questions.

For external-benchmark readiness, we also ported FloodSQL-Bench (Liu et al., 2025a) to a local PostgreSQL/PostGIS schema. The local port contains 443 benchmark questions over 10 flood-risk, social-vulnerability, infrastructure, and boundary tables; 18 questions are reserved as few-shot examples by the runner. A single-sample stratified smoke evaluation (25 questions per L0–L5 level, 150 questions total) with gemini-2.5-flash produced identical baseline and full EX (65/150, 43.3%). We report this only as an external harness-readiness audit. It is not used as evidence for generalisation because it is a single-family, single-sample, partial-benchmark run.

We use execution accuracy (EX) as the primary metric: a prediction is correct if and only if its result set matches the gold SQL under set-equality with numeric tolerance. We additionally

report 95% Wilson confidence intervals, execution validity (fraction of generated SQL that executes without error), per-difficulty breakdowns, and paired McNemar tests. For repeated stochastic LLM samples in §6, the unit of inference is the question: the cross-family panel uses $N=3$ samples per condition, while the focused gemini-3.5-flash A/B/C diagnostic uses balanced $N=5$ samples per condition. In both cases, we first take majority vote across the repeated samples for each condition and question, then run one exact two-sided McNemar test on the resulting paired per-question table. We do not pool repeated samples as independent observations.

Execution accuracy evaluator. EX compares the result set of predicted SQL against gold SQL under the following rules: (i) row order is ignored via set comparison of normalised tuples; (ii) column order follows the SELECT-list positional order and is significant — permuted columns are counted as errors; (iii) numeric tolerance: two values match when $|a - b| \leq 10^{-3}$ (absolute) or $|a - b| / \max(|a|, |b|) \leq 10^{-3}$ (relative); (iv) NULL comparison: NULL equals NULL, and NULL differs from any other value; (v) floats are rounded to three decimal places before set comparison, to damp SQL-engine-specific rounding; (vi) LIMIT: if the gold specifies LIMIT N , the predicted result is treated as-is and the comparison is applied over at most N rows; (vii) duplicate rows (multiplicity) are preserved — set-equality is multiset-equality, not distinct-row equality. The evaluator is a single Python function (`compare_results()`) in the companion repository; all 85 Spatial questions produced by gold SQL return at most 10 000 rows, keeping evaluator cost bounded.

For the framework-level tracks, we compare two pipeline configurations:

- Baseline: direct LLM generation (Gemini 2.5 Flash) with schema dump only, no semantic grounding.
- Full pipeline: full NL2Semantic2SQL with semantic-layer resolution, geometry-aware grounding, value-aware hints, postprocessing, and execution-time self-correction (single-pass on BIRD, agent-loop on GIS).

We additionally compare against DIN-SQL (Pourreza and Rafiei, 2023) as an external prompting baseline on the framework-level tracks. These supporting configurations share the same database backend and execution-comparison harness. The cross-family grounding-effect study in §6 varies the LLM family by design and keeps the A/B prompt conditions, benchmark, evaluator, and question-level test convention fixed. Experimental configurations are detailed in Supplement S1.

5 Baseline Framework Evaluation

Scope note. This section gives compact framework-level context only. It is separate from the primary cross-family grounding-effect study in §6; detailed framework, BIRD, DIN-SQL, token-cost, and case-study tables are retained in Supplement S1. The +32.50 pp Robustness effect reported for gemini-3.5-flash in the cross-family table is the $N=3$ CQ-125 A/B panel value; the focused balanced $N=5$ A/B/C diagnostic reports +27.50 pp for the same family and subset. Neither number is the earlier 15-question pilot noted below.

5.1 Supporting framework results

In the earlier agent-loop GIS run, the best observed framework sample reaches Spatial EX 0.706 on the 85-question set with a gemini-2.5-flash backbone, versus a schema-only baseline of 0.518. This best-sample result is descriptive and is not used as primary inferential evidence. On the 15-question Robustness pilot, the framework reaches 7/7 on discordant pairs ($p = 0.0156$ exact). On BIRD

held-out 150q, the Full pipeline reaches EX 0.507 vs. DIN-SQL 0.440 ($p = 0.0755$, marginal). On the 40-question Robustness suite, the gap concentrates on safety/refusal categories where DIN-SQL has no built-in handling (Full 0.975 vs. DIN-SQL 0.275). These supporting results motivate the framework context; the main CQ-125 claims use the question-level majority-vote protocol in §6.

5.2 Gemma4 model-size sweep on a fixed local host

As a deployment-oriented probe, we ran five Gemma4 model scales on one fixed local Ollama serving node using CQ-125 and the same baseline-vs-full framework configurations. The source table is provided as `gemma4_host228_scale_sweep_summary.csv`. The serving metadata were queried from the same node via the Ollama `/api/tags` and `/api/version` endpoints: Ollama version 0.30.5; GGUF format; Q4_K_M quantisation for all five Gemma4 models; reported parameter sizes of 5.1B, 8.0B, 11.9B, 25.8B, and 31.3B. The semantic retrieval backend on the same node used `nomic-embed-text-v2-moe`. The full model-metadata source table is distributed as `host228_ollama_model_metadata.csv`. This probe is descriptive: it is not an inferential cross-family test and is not used in the primary majority-vote McNemar reductions.

In this fixed-host summary, full-pipeline EX ranged from 80/125 to 114/125, compared with 37/125 to 71/125 for schema-only generation. The Gemma4:31b full run had the highest observed EX (114/125, 20.04 min), while Gemma4:26b was one question lower (113/125, 12.50 min). These numbers are reported only to illustrate an accuracy-runtime deployment trade-off. Because several host228 full-pipeline runs existed for some scales, the CSV identifies the selected full rows as best-observed engineering summaries rather than stochastic significance evidence; the selection procedure precludes inferential comparison across model sizes.

5.3 Reproducibility

All commands and artefacts to regenerate the previously released framework-level reductions are in the anonymised reproducibility repository (<https://anonymous.4open.science/r/nl2geosql-reproduction-36D6/>; reviewer runtime ≈ 90 s with zero LLM API and zero PostgreSQL dependency). These checks reproduce tables from frozen SQL-output and execution-result records rather than re-executing queries against the withheld operational database rows. The cross-family analysis in §6 is regenerated from the same repository. The newer Gemma4 deployment probe is reported from frozen local run summaries; its source-table summary is included with this manuscript source as `gemma4_host228_scale_sweep_summary.csv` and is not used in the primary statistical reductions.

6 Subset-Decomposed Grounding Effects in NL2GeoSQL

6.1 Phenomenon: grounding effects depend on subset and model family

Schema-aware grounding context is conventionally expected to reduce hallucination and improve schema linking in text-to-SQL pipelines. In PostGIS NL2GeoSQL, however, the same intervention touches several distinct behaviours: spatial predicate selection, geometry measurement, join construction, output-shape control, refusal, and bounded execution. We therefore evaluate grounding by subset rather than by aggregate accuracy alone.

The primary panel analysis is the 11-family $B - A$ comparison in Table 7. Under question-level majority vote across $N = 3$ samples, grounding improves the Robustness subset for all 11 families by point estimate. On the Spatial subset, the effect is heterogeneous: most families improve, but

gemini-3.5-flash has the most negative panel estimate (-10.59 pp, $p = 0.136$) and gemini-3.1-pro-preview is near zero negative (-2.35 pp, $p = 0.845$). This panel result establishes heterogeneity rather than a significant family-level regression.

A focused three-condition analysis of gemini-3.5-flash, now balanced at $N = 5$ for no-grounding (A), grounding-only (B), and grounding plus minimal-modification (C), gives the same qualitative but more conservative diagnostic signal. Comparing A with B, the 85-question Spatial subset changes by -10.59 pp ($b/c = 19/10$, exact two-sided McNemar $p = 0.136$), while the 40-question Robustness subset changes by $+27.50$ pp ($p = 0.0010$). The aggregate change is only $+1.60$ pp ($p = 0.875$), which hides the opposite subset directions. We therefore treat the negative Spatial tail as an empirical diagnostic observation rather than a statistically established family-level regression.

Statistical convention. Unless otherwise stated, all paired tests in this section use question-level majority vote across the N stochastic samples followed by a single McNemar exact two-sided test on the resulting per-question table. This avoids the pseudoreplication that would arise from pooling repeated stochastic observations of the same question. The convention is applied uniformly to the three-condition analysis and to the cross-family panel.

6.2 Three-factor decomposition

We decompose the gain space using three conditions per question, paired by sample index:

- A: no-grounding — bare schema dump, no semantic-layer hints. ($N=5$.)
- B: grounding only — standard grounding context (alias hints, value-enum, large-table flags), no profile patch. ($N=5$.)
- C: grounding + minimal-modification (mini-mod) — grounding plus seven no unsolicited X rules targeting over-engineering. ($N=5$.)

Subset	A: no-grounding	B: grounding only	C: grounding + mini-mod
Overall (125q)	53.6%	55.2%	60.8%
Robustness (40q)	65.0%	92.5%	92.5%
Spatial (85q)	48.2%	37.6%	45.9%
Easy (24q)	66.7%	54.2%	66.7%
Medium (36q)	52.8%	38.9%	44.4%
Hard (25q)	24.0%	20.0%	28.0%

Table 5: Question-level majority-vote execution accuracy across $N=5$ samples for gemini-3.5-flash on CQ-125 under three conditions. Bold marks the highest cell per subset. The headline observation is that on the Spatial subset, the no-grounding condition (A: 48.2%) outperforms both grounding-only (B: 37.6%) and grounding+mini-mod (C: 45.9%); on the Robustness subset the order reverses sharply (A: 65.0% < B/C: 92.5%). The aggregate at C is therefore a weighted compromise, not a uniformly Pareto-better operating point.

Paired Δ decomposition. Table 6 reports paired Δ per cell under the question-level majority-vote McNemar convention. The grounding effect ($B - A$) is large and significant on Robustness ($+27.50$ pp, $p=0.0010$), and it is negative but non-significant on Spatial in this focused analysis (-10.59 pp, $p=0.136$). The mini-mod patch effect ($C - B$) recovers $+8.24$ pp on Spatial but does not reach

$\alpha=0.05$ ($p=0.065$, marginal). The total mini-mod-against-baseline effect ($C - A$) is non-significant on Spatial (-2.35 pp, $p=0.84$). Mini-mod therefore remains a candidate partial mitigation, not a statistically validated recovery.

Subset	Grounding ($B - A$)	Mini-mod patch ($C - B$)	Total ($C - A$)
Overall (125q)	+1.60 pp ($p=0.87$)	+5.60 pp ($p=0.092$)	+7.20 pp ($p=0.19$)
Robustness (40q)	+27.50 pp ($p=0.0010$)	+0.00 pp ($p=1.00$)	+27.50 pp ($p=0.0034$)
Spatial (85q)	-10.59 pp ($p=0.136$, n.s.)	+8.24 pp ($p=0.065$, marginal)	-2.35 pp ($p=0.84$, n.s.)
Easy (24q)	-12.50 pp ($p=0.51$)	+12.50 pp ($p=0.38$)	+0.00 pp ($p=1.00$)
Medium (36q)	-13.89 pp ($p=0.27$)	+5.56 pp ($p=0.63$)	-8.33 pp ($p=0.55$)
Hard (25q)	-4.00 pp ($p=1.00$)	+8.00 pp ($p=0.50$)	+4.00 pp ($p=1.00$)

Table 6: Paired Δ ($\pm$ McNemar exact two-sided p) under question-level majority-vote across $N=5$ samples. Bold marks results significant at $p<0.05$. The Spatial grounding effect in this focused gemini-3.5-flash analysis remains negative ($B - A = -10.59$ pp), but it is non-significant. The mini-mod patch effect on Spatial ($C - B = +8.24$ pp) is marginal; the total $C - A$ is non-significant.

Why de-pooled tests matter here. A naive pooled test would treat 85 Spatial questions $\times N=5$ samples as 425 independent paired observations, drastically inflating the effective sample size. Same-question samples are not independent: they share question difficulty, schema-mapping ambiguity, and gold-SQL idiosyncrasies. The honest design is one paired observation per question (the within-sample majority vote), giving 85 Spatial observations. The price is statistical power; the gain is a defensible p -value.

Mixture sensitivity. The focused gemini-3.5-flash A/B contrast also shows why aggregate EX depends on benchmark composition. Let w be the Spatial share of a mixed benchmark. The expected aggregate grounding effect is

$$\Delta(w) = w(-10.59) + (1 - w)(27.50).$$

The effect changes sign at $w \approx 0.722$. CQ-125 has $w = 85/125 = 0.68$, so its aggregate effect is close to zero (+1.60 pp) even though the subset effects are opposite-signed. A robustness-heavy benchmark would make grounding look clearly positive, while a more spatial-heavy benchmark could make the same A/B contrast look negative. This analytic sensitivity check does not add new data; it makes the dependence on benchmark composition explicit.

6.3 Cross-family evidence: heterogeneous grounding effects across 11 LLMs

We extend the analysis to ten additional LLMs whose CQ-125 $N=3$ baseline-vs-full data are available from prior batch runs. None uses mini-mod; we report only $B - A$ paired Δ under the same question-level majority-vote McNemar convention. This is the primary cross-family evidence in the paper.

Multiplicity. The star markers in Table 7 are unadjusted per-cell p -values retained to show the panel pattern. Treating the 11 Spatial and 11 Robustness tests as two exploratory comparison sets, Holm correction at $\alpha=0.05$ leaves 3 Spatial cells and 9 Robustness cells significant; Benjamini-Hochberg FDR control at $q=0.05$ leaves 6 Spatial cells and 9 Robustness cells significant. We therefore read counts of significant family cells descriptively. The confirmatory claim is the subset-conditional heterogeneity pattern under the de-pooled question-level convention, not a corrected discovery claim for every individual family.

Family	Spatial (85q)			Robust (40q)		
	A mean %	B mean %	MV Δ (pp)	A mean %	B mean %	MV Δ (pp)
gemini-2.5-flash	54.9	68.6	+12.94 *	40.0	97.5	+57.50 **
gemini-2.5-pro	57.6	72.5	+18.82 **	61.7	91.7	+30.00 **
gemini-3.1-flash-lite-preview	50.2	67.8	+16.47 **	70.0	85.0	+15.00 (n.s.)
gemini-3.1-pro-preview	65.1	63.5	-2.35 (n.s.)	63.3	99.2	+35.00 **
gemini-3.5-flash	48.6	36.5	- 10.59 (p=0.14)	58.3	89.2	+32.50 **
deepseek-v4-flash	43.1	62.8	+23.53 **	28.3	81.7	+62.50 **
deepseek-v4-pro	48.2	61.6	+15.29 (p=0.05)	28.3	90.0	+65.00 **
qwen3.6-flash	35.3	49.8	+12.94 (p=0.06)	45.8	72.5	+27.50 **
qwen3.6-plus	50.2	65.1	+12.94 *	66.7	91.7	+30.00 **
qwen3.7-max	44.3	64.3	+17.65 **	73.3	94.2	+22.50 *
gemma-4-31b-it	44.7	57.3	+11.76 *	89.2	95.8	+12.50 (p=0.06)

Table 7: Cross-family question-level majority-vote Δ on CQ-125 ($N=3$ samples per family). A and B columns report per-sample mean execution accuracy across the three samples for readability; MV Δ is computed after question-level majority vote as $MV(B) - MV(A)$ in percentage points. Significance is the paired exact two-sided McNemar test on the per-question majority-vote table. Significance markers are unadjusted per-cell values: * $p<0.05$, ** $p<0.01$. Bold-italic marks the most negative panel point estimate (gemini-3.5-flash, $p=0.136$). Of 11 families, 7 show unadjusted significant positive Spatial Δ , 2 marginal positive (p between 0.05 and 0.10), and 2 non-significant negative. Robustness Δ is positive for every family, with nine unadjusted significant, one marginal, and one non-significant positive cell.

What the table establishes and what it does not. The robust finding is that grounding produces a heterogeneous family-conditional effect on the Spatial subset. The point estimates range from -10.59 pp (gemini-3.5-flash) to +23.53 pp (deepseek-v4-flash); the difference between these tails is 34.12 pp. The Robustness subset, in contrast, has positive point estimates across all 11 families (+12.50 to +65.00 pp).

The table does not establish a stable negative model-family property for gemini-3.5-flash. Its panel Spatial estimate is the most negative in magnitude, but it is not significant at $\alpha=0.05$ ($p = 0.136$). The balanced $N = 5$ three-condition check gives the same Spatial point estimate under the majority-vote convention, so it supports the negative-tail diagnosis without converting it into a confirmatory claim.

Aggregate metrics are insufficient. The single most important point from Table 7 is that aggregate execution accuracy on a benchmark mixing Robustness and Spatial questions is a poor proxy for the underlying interaction. A reader who consulted only an Overall Δ column could conclude that grounding adds a small positive effect across the panel. The subset decomposition shows that the positive aggregate can be driven by Robustness gains while Spatial gains vary strongly by family. The methodological contribution is therefore that subset decomposition with question-level paired tests is necessary for geospatial NL2SQL benchmarks of this type.

6.4 Mini-mod profile and its (limited) effect

The mini-mod profile prepends seven no unsolicited X rules to the system prompt of gemini-3.5-flash only:

1. No unsolicited GROUP BY unless explicit grouping language is present;
2. No unsolicited aggregation (SUM/AVG/MAX/MIN/COUNT);

3. No unsolicited ST_Union;
4. No unsolicited ROUND;
5. No unsolicited nested subqueries; flat SELECT preferred;
6. No unsolicited ST_Transform;
7. Treat grounding hints as reference, not checklist.

These rules target the SQL-elaboration patterns that dominate the regression diff between A and B on gemini-3.5-flash: per-record analysis at $N=3$ shows nested subquery wrapping (29.7% of regressions), redundant WHERE (17.2%), unsolicited GROUP BY (14.1%), and ST_Union pre-aggregation (4.7%) as the largest contributors.

Empirical effect under de-pooled statistics. Per Table 6, mini-mod does not achieve a statistically significant patch effect at $\alpha=0.05$. The point estimate $C - B = +8.24$ pp on Spatial is marginal ($p=0.065$), and the total $C - A = -2.35$ pp is not significant ($p=0.84$). We therefore characterise mini-mod conservatively: a candidate prompt-level partial mitigation, not a validated production technique. The patch’s specificity to gemini-3.5-flash and the fact that no-grounding still outperforms grounding plus mini-mod on Spatial ($A=48.2\%$ vs. $C=45.9\%$) indicate that prompt-level patching is not yet a sufficient abstraction for this family.

6.5 Design implication: intent-aware grounding gating as upper-bound analysis

Table 5 suggests a possible design direction, not a validated production policy. On gemini-3.5-flash, the best observed static choice differs by subset: Robustness-class queries gain large EX from grounding (cells B/C, 92.5%), while Spatial-class queries have the highest EX under no-grounding (cell A, 48.2%). If a hypothetical perfect intent classifier could route each question correctly, the resulting upper-bound EX on CQ-125 would be $\frac{40 \cdot 92.5 + 85 \cdot 48.2}{125} \approx 62.4\%$, exceeding any single static policy ($A=53.6\%$, $B=55.2\%$, $C=60.8\%$).

What we report and what we do not. This calculation is an upper-bound theoretical analysis. We do not report end-to-end EX under a real classifier in the present manuscript. A complete intent-aware-gating evaluation requires (i) classifier accuracy / precision / recall on CQ-125, (ii) end-to-end EX after classifier errors, and (iii) latency and token-cost overhead from the additional classification step. The intent-router implementation referenced from our companion repository operates over a coarse spatial-vs-robustness label, but its empirical accuracy on CQ-125 has not been validated under the same evaluation harness as the grounding cells. The contribution of this subsection is therefore the upper-bound number and the qualitative implication that subset-conditional grounding is a plausible policy class for this family.

For all other tested families, plain grounding remains the empirically best static policy at $N=3$; the gating logic is family-conditional.

6.6 Methodological lesson

We propose a preregistration-ready analysis template for grounding-effect studies on geospatial NL2SQL benchmarks of mixed Spatial–Robustness composition:

1. Data manifest first. Enumerate every $N \geq 3$ JSONL in the corpus before constructing any analysis table.

2. Three conditions per family (A no-grounding, B grounding-only, C grounding+patch) at $N \geq 5$.
3. Per condition: Overall, Robustness vs. Spatial, Easy/Medium/Hard, plus top- k category breakdowns where $n_q \geq 3$.
4. Per subset, use a question-level paired test. The honest paired test for repeated stochastic LLM samples is question-level majority vote across N samples followed by a single McNemar exact two-sided test on the per-question table; pooling repeated observations across samples violates independence and inflates significance.
5. Report decomposition $B - A$, $C - B$, $C - A$ separately; any narrative must attribute the gain to one of these.
6. State the multiplicity policy before counting significant family cells; if correction is exploratory rather than confirmatory, say so explicitly.
7. Pre-specify at least three competing interpretations of the expected result, including null and negative.

Every number in Tables 5–7 is regenerable from the released JSONL via the offline analysis scripts shipped in the companion repository.

6.7 Limitations

1. Finite-sample statistical regime and multiplicity. The primary cross-family panel remains at $N=3$ paired samples per question for comparability across all 11 families, while the focused gemini-3.5-flash A/B/C analysis is balanced at $N=5$. The balanced check retains a negative Spatial point estimate but no longer reaches $\alpha=0.05$ under question-level majority-vote McNemar testing. The table-level star markers are unadjusted; after Holm correction within the 11-family Spatial comparison set, three positive Spatial cells remain significant, whereas the gemini-3.5-flash negative tail remains non-significant. We report it as a diagnostic negative-tail observation, not an established model-family property.
2. Single benchmark. CQ-125 is a Chongqing-grounded geospatial benchmark with 85:40 spatial:robustness mixture. Replication on benchmarks of different composition (e.g. pure-spatial GeoSQL-Eval, pure-warehouse) is required before claiming the phenomenon generalises.
3. Mini-mod is a candidate patch. The balanced $N=5$ patch effect on Spatial is marginal ($p=0.065$); we do not claim it as a validated technique.
4. Intent-aware gating is presented as upper-bound analysis. A classifier-validated end-to-end EX evaluation is not reported in this manuscript. The upper-bound calculation is not a production-deployable claim.
5. Subset of LLMs surveyed. The 11-family panel includes only models with API access during the evaluation window. Replication on additional model providers is required before treating the observed distribution of effects as representative of the broader model ecosystem.

6.8 Reproducibility

All numerical results in this section — the three-factor decomposition (Tables 5–6) and the cross-family Table 7 — are regenerable offline from frozen evaluation records with zero LLM API calls, zero PostgreSQL dependency, and zero GPU requirement. This supports reproducibility of the statistical reductions from frozen SQL-output and execution-result records; it is not a

raw-row re-execution package because the operational database rows are withheld. The reviewer-accessible anonymised reproduction package is available at <https://anonymous.4open.science/r/nl2geosql-reproduction-36D6/> for double-anonymised review. It ships frozen JSONL execution traces, benchmark question metadata, prompt profiles, and offline analysis scripts for all eleven LLM families. Each primary panel run record is keyed by (family, sample_index, mode, qid) and can be reduced by the question-level majority-vote convention used in this manuscript. The package also releases the schema-level materials needed to audit grounding without exposing sensitive row-level records: a structure-only schema.sql dump, table/column metadata, geometry-column metadata, SRID information, semantic-layer configuration, the minimal-modification prompt profile, and the upper-bound intent-gate decision template. Embeddings used during semantic resolution are precomputed and cached in the JSONL traces, so reviewers do not need to invoke any embedding model. The repository includes a CI-style `verify_tables.py` check for one frozen cross-family table; the majority-vote reductions reported here are computed from the same JSONL records under the de-pooled convention stated above.

7 Discussion

7.1 A Spatial–Robustness inversion in one fast-model family

Per-record diff between A (no grounding) and B (grounding only) on gemini-3.5-flash suggests that many regressions involve SQL elaboration the question did not ask for. Among 91 distinct *A*-pass / *B*-fail pairs across $N = 3$ samples (pooled for diagnostic, not for inference), the four largest patterns are nested subquery wrapping (29.7%), redundant WHERE clauses (17.2%), unsolicited GROUP BY (14.1%), and ST_Union pre-aggregation before ST_Centroid/ST_Area (4.7%). Each can shift the result-set shape away from the gold answer even when the model has identified the relevant table and spatial concept.

A geospatial cognitive-load reading. One GIScience interpretation is that grounding context, by exposing more table-and-column metadata than the question itself uses, may encourage gemini-3.5-flash to compose an “explanatory” SQL statement that incorporates that metadata. Such statements can aggregate across a parent table, join to context tables, or project additional descriptive columns while no longer matching the gold result. We refer to this diagnostic pattern as geospatial over-engineering: prompt-side spatial evidence (geometry types, SRID ranges, predicate taxonomies, alias hierarchies) may shift from reference material to checklist material for some model families. The Spatial subset is more exposed to this failure mode because spatial questions admit a wider space of plausible-but-wrong elaborations; Robustness questions are more structurally constrained by refusal, bounded preview, and schema validation.

A second possible factor is baseline headroom. Several families that gain from grounding on the Spatial subset have weaker no-grounding EX — for example deepseek-v4-flash (43%), qwen3.6-flash (35%), and qwen3.7-max (44%) — so the marginal information from grounding may outweigh the over-elaboration risk. gemini-3.5-flash starts at 48.6%, leaving less room for grounding to add schema-linking value. A third possibility concerns instruction tuning in fast model tiers. Models tuned strongly for refusal and safety behaviour may benefit on Robustness while becoming less able to ignore optional context on ordinary spatial queries. We do not have access to model internals, so this remains an interpretation rather than a mechanistic claim.

7.2 Why subset decomposition is not optional

The CQ-125 benchmark mixes 85 spatial-query questions and 40 robustness questions. For the same family, the focused three-condition analysis gives opposite-signed subset effects for grounding-only versus no-grounding: -10.59 pp on Spatial and $+27.50$ pp on Robustness. Any aggregate metric over a benchmark with this kind of mixture can obscure the interaction. This lesson is directly relevant to GIScience evaluation: spatial predicate translation, metric measurement, bounded preview, and refusal are different behaviours and should not be collapsed into a single headline without subset reporting.

7.3 Model-size scaling is a deployment question, not only a leaderboard question

The fixed-host Gemma4 sweep is included only as a deployment probe. Under the same CQ-125 benchmark and local Ollama host, the full pipeline improved all five tested Gemma4 scales by descriptive point estimate. The largest model, Gemma4:31b, had the highest observed full-pipeline EX (114/125). The Gemma4:26b run was one question lower (113/125) and used 12.50 minutes rather than 20.04 minutes.

This result should not be interpreted as statistical evidence about model families. It uses single-host execution summaries, not the $N=3$ majority-vote protocol used for the cross-family grounding-effect panel. Its value is practical: semantic-layer grounding may narrow the gap between mid-size and larger local models, so deployment choice can require an accuracy-runtime trade-off rather than a largest-model rule.

7.4 Limitations

1. Finite-sample regime; multiple-comparison correction. All primary cross-family panel cells are at $N=3$ paired samples per question. The focused gemini-3.5-flash A/B/C check is balanced at $N=5$ and retains a negative Spatial point estimate, but it remains non-significant ($p = 0.136$). Cross-family table markers are unadjusted per-cell p-values used descriptively. Holm correction within the 11 Spatial family tests leaves three positive cells significant and does not make the gemini-3.5-flash negative tail significant; Holm correction within the 11 Robustness tests leaves nine positive cells significant. We label the negative tail a diagnostic observation, not a confirmed model-family property.
2. Single primary benchmark. CQ-125 has a 85:40 spatial:robustness mixture. We have ported FloodSQL-Bench to the same PostgreSQL/PostGIS evaluation harness and completed a single gemini-2.5-flash stratified smoke run, but that run is not a full external replication. Confirmation requires a pre-specified multi-family, $N \geq 3$ run on benchmarks of different composition, including pure-spatial GeoSQL-Eval (Hou et al., 2026), FloodSQL-Bench (Liu et al., 2025a), and a pure-warehouse BIRD held-out set.
3. Mini-mod is a candidate patch. The balanced $N=5$ patch effect on Spatial is marginal ($p = 0.065$) and we do not claim it as a validated technique.
4. Intent-aware gating is upper-bound analysis. A classifier-validated end-to-end EX evaluation, including classifier P/R/F1 on CQ-125 and end-to-end EX after misrouting, is not reported in this manuscript and is identified as immediate follow-up.
5. Subset of LLMs surveyed. The 11-family panel includes only models with API access during the evaluation window. Replication on additional model providers is required before treating the observed distribution of effects as representative of the broader model ecosystem.

6. Gemma4 scale sweep is a deployment probe. The five-size Gemma4 sweep is reported to expose a practical accuracy–runtime trade-off on a fixed local host. It is not a stochastic $N=3$ family-level inference panel, and its best-observed full-pipeline rows should not be interpreted as statistical significance evidence.
7. Question phrasing. Four of 125 questions admit superset-column readings. They are two Easy items and two Medium items in CQ-125. A relaxed col-count ablation does not change the qualitative conclusion.

7.5 Future work

(i) External replication of the balanced A/B/C diagnostic on benchmarks with different composition, including a pure-spatial PostGIS benchmark, FloodSQL-Bench, and a pure-warehouse benchmark. (ii) End-to-end intent-aware-gating evaluation under a real classifier with reported P/R/F1 and post-misrouting EX. (iii) Extension of the $N \geq 5$ three-condition protocol beyond gemini-3.5-flash to additional LLM families. (iv) Inclusion of OpenAI and Anthropic families to test whether the family-conditional pattern is specific to the currently available model panel. (v) Mechanistic study of the over-engineering interpretation through controlled perturbation of grounding-context length and content.

8 Conclusion

We report a question-level paired-McNemar evaluation of schema-aware grounding across 11 LLM families on the 125-question PostGIS NL2GeoSQL benchmark CQ-125. The grounding effect is heterogeneous and family-conditional. On the 85-question Spatial subset, point estimates range from -10.59 pp (gemini-3.5-flash, $p = 0.136$) to $+23.53$ pp (deepseek-v4-flash, $p < 0.001$). On the 40-question Robustness subset, all 11 families improve in point estimate. A balanced $N=5$ three-condition analysis of gemini-3.5-flash retains the negative Spatial point estimate for grounding-only versus no-grounding (-10.59 pp, $p = 0.136$), but does not convert it into a significant family-level regression. The methodological conclusion is stronger: aggregate execution accuracy is insufficient on mixed Spatial–Robustness benchmarks, and pooled McNemar tests over repeated stochastic samples inflate significance. We recommend subset-decomposed reporting and question-level majority-vote McNemar tests as the default evaluation protocol for grounding studies in geospatial NL2SQL.

Author Contributions

Ning Zhou conceived the study, designed the benchmark and semantic-layer evaluation protocol, implemented the analysis scripts, ran the experiments, verified the statistical analyses, wrote the manuscript, and approved the final version for submission.

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Competing Interests

The author declares no competing interests.

Ethics Statement

This study uses benchmark questions, schema-level database metadata, generated SQL, and offline execution/evaluation records. It does not report experiments on human participants, patient data, or animal subjects. The underlying operational database rows are withheld because they contain sensitive local data.

Use of Artificial Intelligence Tools

Large language models are the objects of evaluation in this study, and their use in the experimental protocol is described in the manuscript. Claude Code and OpenAI Codex were also used for editorial support, language polishing, format checking, and consistency review during manuscript preparation. The author checked and revised all AI-assisted text, verified the accuracy of the manuscript and references, and takes full responsibility for the final content. No AI tool is listed as an author.

Data and Code Availability Statement

An anonymised reproduction package for double-anonymised review is available at <https://anonymous.4open.science/r/nl2geosql-reproduction-36D6/>. The package contains offline analysis scripts, schema manifests, semantic-layer configuration, benchmark question metadata, prompt profiles, and pre-computed SQL-output and execution-result records that support offline regeneration of the primary CQ-125 grounding-effect analyses and the previously released framework-level tables. These checks require no LLM API calls, database access, GPU resources, or access to sensitive raw records. This is reduction-level reproducibility from frozen execution records, not independent raw-row re-execution. The balanced gemini-3.5-flash N=5 A/B/C diagnostic summary, the fixed-host Gemma4 deployment sweep, and the FloodSQL-Bench smoke audit added in this revision are provided as source-table summaries with the manuscript source. The underlying operational database rows are not released because they contain sensitive local data. Instead, we release the schema-level information required for auditability: table/column structure, geometry-column metadata, SRID information, and semantic grounding rules. The repository license file specifies MIT licensing for code and CC-BY 4.0 licensing for benchmark questions and frozen LLM evaluation outputs under data/. After acceptance, the anonymised review link will be replaced by a permanent archival repository with the same file layout and explicit license text.

Item	Availability			Reproducibility role
Primary traces	CQ-125	Anonymised review repository		Regenerate the subset-decomposed majority-vote tables and McNemar tests without LLM or database access.
Schema and grounding metadata		Anonymised review repository		Audit table structure, geometry columns, SRID information, and semantic-layer rules without exposing row-level records.
Gemini 3.5 diagnostic	A/B/C	Manuscript source CSV		Audit the balanced N=5 majority-vote summary used in the focused three-condition analysis.
Gemma4 deployment probe		Manuscript source CSVs		Audit the descriptive fixed-host scale-sweep summary and Ollama model metadata; not used in primary statistical reductions.
FloodSQL-Bench smoke audit		Manuscript source CSV		Audit external-harness readiness; not used as generalisation evidence.
Software/data license		Repository LICENSE		Code released under MIT; benchmark questions and frozen LLM outputs under CC-BY 4.0.
Operational row data		Withheld		Not released because the database contains sensitive local records.
Permanent archive		After acceptance		Replace the anonymised review link with a durable repository and explicit software/data license.

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